

Jester Gesture Recognition: Single-Frame 2D CNN vs Clip-Based 3D CNN

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Abstract

In this project, gesture recognition models were developed on the Jester dataset, which consists of short clips of video of 27 hand gesture classes. Our main goal is to maximize validation accuracy while working under limited hardware resources.

As a baseline, a 2D ResNet18 model was implemented. It classified videos using a single frame. An improved pretrained 3D ResNet model was created, which processes short clips of consecutive frames, including temporal information.

Experimental results of the validation set indicate that the improved model outperforms the baseline, achieving 86%, which is 35% higher than the 51% baseline model. Analysis of confusion matrices shows that the 3D model reduces confusion between gestures that are similar in static pose but are different in motion direction. It indicates that the improved model still has some errors, i.e. with gestures in opposite directions. A major limitation is the use of short video clips and a reduced training set. In future works, models could receive longer temporal inputs and training on larger datasets.

Introduction and Related Work

Problem Definition & Motivation

Compared to image classification, gesture classification is more difficult as its label is defined as a motion over a specific time frame, instead of a static pose. Gestures always include hands but differ in direction, sequence over time, and changes in hand position across frames. Thus, when a model looks at only one frame, it has to "guess" the intended gesture based on that specific frame.

In this project, we use the Jester dataset, a large collection of short hand gesture videos designed for action/gesture recognition, to classify 27 hand-gesture classes [3]. In this dataset, there are many gestures that share some nearly identical states at some frames, such as swiping left/right/up/down, that can be considered a group for their shared movements, which makes it a useful dataset for comparison of static and dynamic movements.

Constraint Definition

The constraint in this project is to maximize validation accuracy while working under limited hardware resources. Video models are more expensive than image models, as they process multiple frames

per sample, which increases memory and computation. Specifically, we use a short clip length, train for relatively few epochs, and use pretrained models from related tasks as a strong starting point.

Related Work

A baseline model for video recognition is to do image classification by sampling one or a few frames of each sample and using a 2D CNN. For image classification using 2D CNN, residual networks (ResNets) are useful due to their strong performance [1]. However, using these methods has the limitation of not capturing motion, making them underperform when the motion over time defines the different classes, and not the appearance.

Many methods use 3D CNNs, which learn both spatial and temporal features from clips of frames. A study about this shows that 3D CNNs can capture motion patterns and improve action recognition, especially for direction-dependent classes [4]. Another study factorises these 3D convolutions into separate spatial and temporal components, making the model more efficient, easier to train, and able to capture motion over time [2]. These support the idea that modelling time can be essential for the task of hand gesture recognition.

Finally, using a pretrained model is important considering limited compute and training time. Instead of learning spatial and temporal features from the start, using a pretrained model can accelerate convergence and improve accuracy.

Methodology

Dataset & Preprocessing

The Jester dataset was used, where short hand gesture videos were stored as folders of RGB frames together with CSV files assigning every video ID to the corresponding gesture label. Due to a varying number of frames of each video, consistent sampling was required for both the training and evaluation datasets. The model is trained on the provided 20% subset of the dataset to reduce computational cost,

but evaluated on the full validation set, ensuring a proper comparison between models under the chosen constraint. We repeat the training set twice per epoch (`dataset.repeat = 2`) to get more training updates.

For the baseline 2D model, a single frame is sampled from each video. During training, a random frame is selected to increase diversity, while during validation, the middle frame is used for evaluation. Moreover, during training, standard spatial data augmentations were applied, such as random resized crop (112 x 112), rotation, or color jitter.

For the improved 3D model, each sample consists of a clip of $T=16$ consecutive frames. If a video contains fewer than T frames, the last frame is repeated to pad to get length T . During training, a random starting frame is selected, where, for validation, a centered clip is chosen. Spatial data augmentation, such as resize, crop, and color jitter, was also applied independently to each frame in the clip to improve generalization and regularization.

Baseline Model

As a baseline, a 2D convolutional model was implemented that classifies each video using one RGB frame, without explicit temporal modeling. For each training sample, a random frame from the video is chosen, whereas for validation, the middle frame is used.

The model architecture is ResNet18, initialized without pretrained weights. The last fully connected layer is replaced with a linear layer producing 27 output classes, corresponding to gesture labels in the Jester dataset.

Training is performed using cross-entropy loss with label smoothing and AdamW optimizer. A cosine annealing learning rate is applied to improve convergence over training [5]. The model is trained for 20 epochs with a batch size of 32, and is used as a baseline as it relies entirely on spatial appearance, ignoring motion across frames.

Improved Model

For the improved model, to incorporate temporal information, the model was enhanced to process a video of consecutive frames rather than a single image for every video. Each input consists of a T -frame clip. During training, a random temporal window was chosen for every input, whereas during validation, a centered clip was selected.

The model architecture was improved to a pretrained 3D ResNet (r3d_18), which applies 3D convolutions to gather information about both spatial and temporal features. The last pretrained classification

layer was replaced with a new fully connected layer that produces 27 gesture classes. A dropout layer ($p=0.5$) is added before the final classifier to reduce overfitting.

Training the 3D model is more intensive in memory than the baseline, and therefore, a smaller batch size and a smaller number of epochs are used. The model is optimized with AdamW and cross-entropy loss with label smoothing. Instead of cosine annealing as in the baseline, the OneCycle learning rate schedule was used in the improved model [6].

Due to explicit storing of temporal dynamics, the improved model is able to distinguish gestures that differ in motion direction or temporal ordering even when sharing similar static poses. Spatial augmentations are applied to each frame, disrupting temporal consistency but improving generalization.

Training Protocol

Table 1 summarizes the hyperparameters that differ between the two models. We tried to keep the same hyperparameters as much as possible to make a fair comparison. In both models, AdamW was used as optimizer and cross-entropy with label smoothing (0.1) as loss function. Moreover, for both models, we enabled mixed precision training using autocast and GradScaler to improve the models. After each epoch, models are evaluated on the validation set, and the best checkpoint is selected based on the highest validation accuracy.

Hyperparameter	Baseline (2D ResNet18)	Improved (3D r3d_18)
Epochs	20	10
Batch size	32	8
Learning rate (max)	1×10^{-3}	1×10^{-3}
Weight decay	5×10^{-4}	5×10^{-4}
LR scheduler	Cosine AnnealingLR	OneCycleLR
Scheduler step	once per epoch	once per batch

Table 1: Training hyperparameters and their values for the baseline and improved models.

Evaluation Metrics

The main evaluation metric for the project is validation accuracy, which reflects the goal of maximizing validation accuracy under limited hardware resources.

To gather more insights about model behaviour, the confusion matrices were computed for both models. The metrics correspond to recall per class, which helps in understanding how every gesture is recognized and which classes are most confused. These

matrices allow us to identify errors between gestures that have a similar appearance but different motion patterns.

Training and validation accuracy and loss curves across epochs are also reported to support analyzing convergence behaviour, stability of training, and the existence of overfitting. They also help with the comparison between two models in terms of learning.

Experiment Results & Discussion

Quantitative Results

We compare the 2D baseline model against the 3D improved model. Their performance is evaluated on the full Jester validation set. The baseline model reaches a best validation accuracy of 51%, and the improved model 86%, thus +35% gain.

The accuracy curves, illustrated in Figure 1, show different learning. The baseline model improves gradually across epochs and then plateaus early, which suggests that it is limited by the single-frame inputs. The improved model converges quickly, with the accuracy rapidly increasing in the first epochs and stabilizing in the last ones.

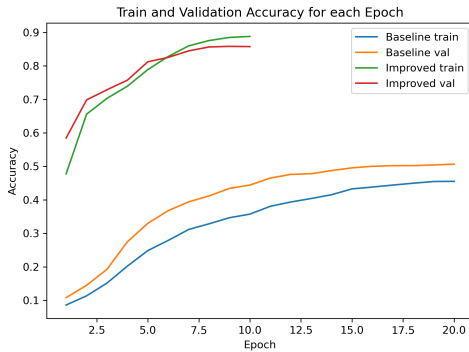


Figure 1: Training and validation accuracy curves for the baseline 2D ResNet18 and the improved 3D r3d_l8 model. The improved model converges quickly and reaches 86% validation accuracy, higher than the validation accuracy of 51% of the baseline model.

Additionally, this is also supported by their loss curves shown in Figure 2. The baseline’s loss decreases but remains relatively high, showing that the model is still unsure between gesture classes, while the improved model’s loss has a sharper loss decrease, showing clearer class separation.

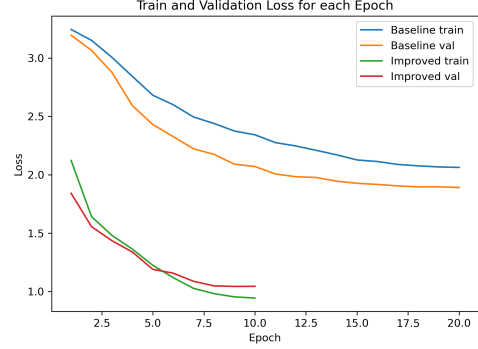


Figure 2: Training and validation loss curves for the baseline 2D ResNet18 and the improved 3D r3d_l8 model. The improved model has a faster loss reduction and stabilizes at a much lower loss than the baseline model.

Confusion Matrix Analysis

Confusion matrices provide insights into which gesture classes are difficult for each model to classify. The confusion matrices of our experiment are row-normalized, so each row represents the recall of each class.

From the baseline model’s confusion matrix illustrated in Figure 3, it is shown that errors happen in gesture groups that include more complex motion directions (i.e., swiping left/right/up/down, sliding two fingers left/right/up/down, zooming in/out), instead of a static pose (i.e., thumb up/down, stop sign).

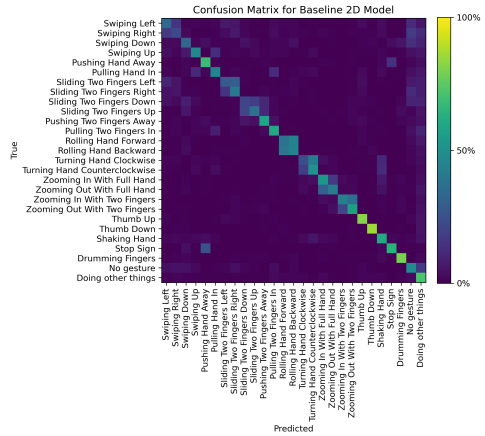


Figure 3: Row-normalized confusion matrix for the baseline 2D ResNet18 evaluated on the validation set. Errors mostly happen between gesture groups with motion directions.

Looking at the improved model’s confusion matrix in Figure 4, it is noticeable that the diagonal

is strengthened, which means that there is a high recall for most classes. This shows that the confusion is reduced for similar gesture groups, but there are some errors for opposite gestures (i.e., turning hand clockwise/counterclockwise).

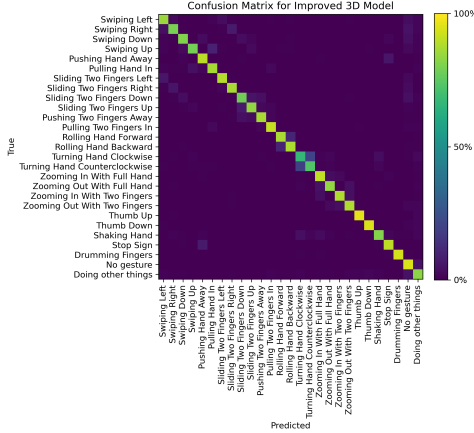


Figure 4: Row-normalized confusion matrix for the improved 3D r3d.18 evaluated on the validation set. Errors mostly happen in opposite-direction pairs.

Qualitative Error Analysis

To enhance the quantitative results, we inspected some individual validation clips to understand where errors occur.

The first example illustrates a limitation of the baseline model, as the baseline predicts "No gesture" instead of "Swiping Right" for that video. On the other hand, the improved model correctly predicts "Swiping Right", as this model takes into account the direction of motion (Figure 5).

The second example shows a case where the improved model also fails. Specifically, both baseline and improved models classified that video as "Turning Hand Clockwise", while the correct label is "Turning Hand Counterclockwise". These classes are more difficult to classify as they are opposite and mainly differ in rotation direction (Figure 5).

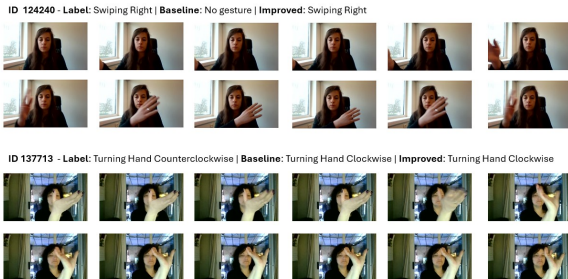


Figure 5: Validation examples selected from misclassified samples. Top clip: "Swiping Right" clip misclassified by the 2D baseline ResNet18 as "No gesture", while the 3D improved r3d.18 classifies it correctly. Bottom clip: "Turning Hand Counterclockwise" clip misclassified as "Turning Hand Clockwise" by both models.

Limitations

This comparison is not completely clean as multiple factors differ between the baseline and improved models, including input representation (single frame vs clip), model type (2D vs 3D), use of pretrained model, the learning rate scheduler, and more. Therefore, the improvement should be connected to the combined effect of these different factors and not a specific component of the improved model. Additionally, training was performed using the provided small training subset, and performance could improve using the full training data and longer training.

Conclusion

In this project, a 2D CNN baseline based on a single frame was compared with a clip-based 3D CNN for gesture recognition on the Jester dataset. The baseline model considered only a spatial feature from one frame per video, whereas the improved model included temporal information by processing sliced videos of consecutive frames.

The results show that the 3D model outperforms the 2D baseline, achieving higher validation accuracy. The model's ability to receive information about motion direction and temporal ordering, and usage of pretrained weights, provided better results. Due to these improvements, the model had strongly initialized weights and converged faster.

While analyzing the confusion matrices, they revealed that the baseline has difficulties distinguishing gestures that have similar static appearances but are different in motion, whereas the improved model almost resolves this confusion, while gestures with opposite directions are observed to cause a problem.

Future work could explore adaptive lengths of the clip, better consistency between temporal augmentations, and training on a larger dataset to improve accuracy and robustness.

Individual Contributions

Both team members contributed equally to this project. We both agreed on the project constraint and found and reviewed related studies. One team member focused more on implementing and tuning the baseline 2D model, while the other focused more on implementing and tuning the improved 3D model. Also, the scripts for experiments and the report writing were split approximately equally. Throughout the project, we reviewed each other's code and writing to ensure consistency.

References

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